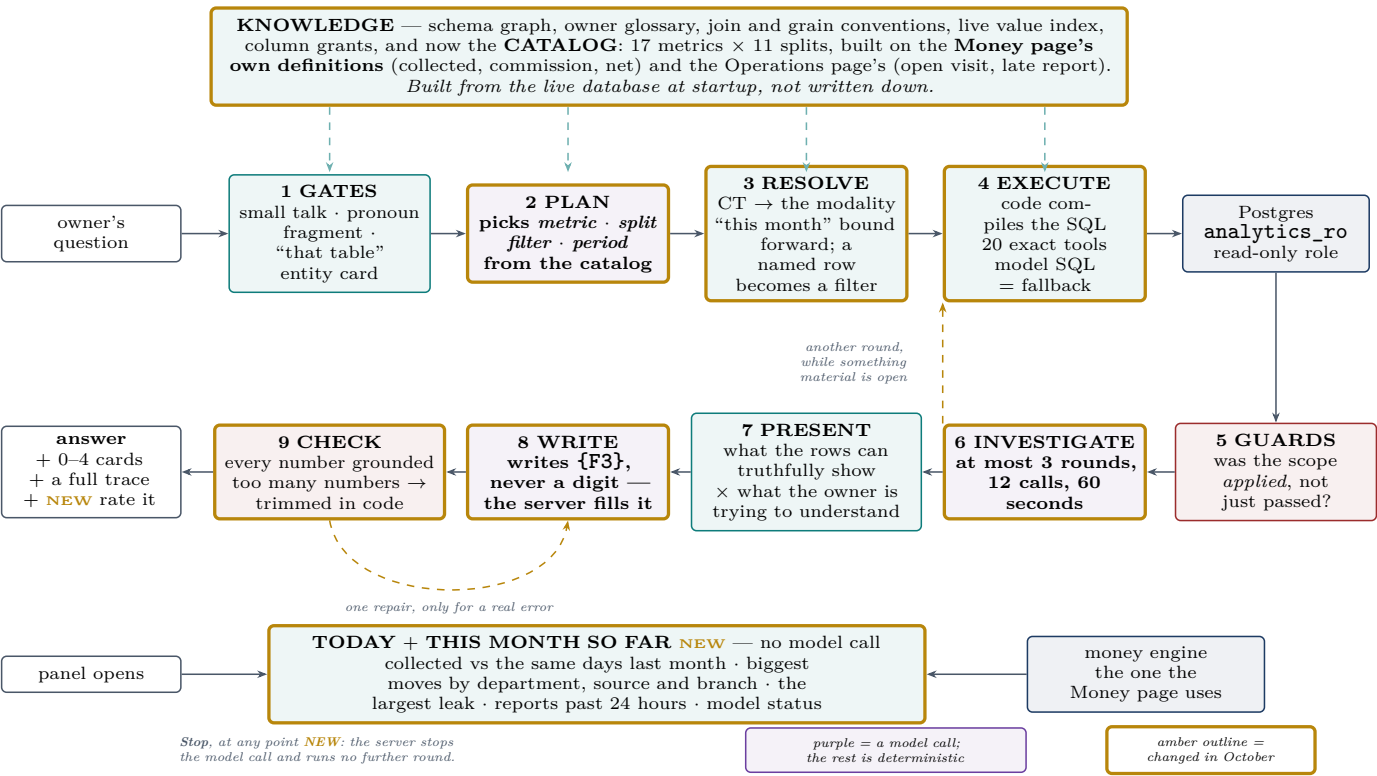


In September Pulse planned an analysis and then *wrote SQL* for half of its steps, typed every figure into its answer by hand, and kept its own definitions of money. In October it stopped writing what it can look up. The model now picks a metric from the same definitions the Money page uses, code writes the SQL, and the model *points at* figures instead of typing them.

How one question flows now



Cost of that: 2 model calls and about 4–6 seconds for a figure (plan, then write). An open question is 2–6 calls. The hard ceiling is 60 seconds and 12 calls, down from 180 seconds, and the answer must say so if it hits it.

What we upgraded

Upgrade	What was wrong	What it does now
Catalog first	The model wrote raw SQL in 50% of steps and 29% of those failed. "give me cost" took 31 calls.	The model picks metric, split, filter and period; code compiles the SQL. Free SQL is only the fallback, now 4% of steps.
The Money page's numbers	Pulse had its own revenue and commission formulas, so its totals disagreed with the Money page.	Collected, commission and net come from the same money engine. Checked equal to the rupee for yesterday, month to date and 30 days.
Figures by reference	The writer typed numbers. 15 of 31 answers needed a second, rewrite call.	The writer writes {F3}; the server fills in the value. A figure can only be one that a step measured.
This month so far	Nothing appeared until the owner asked a question.	A digest card with no model call: collected against the same days last month, the biggest moves, the largest leak and late reports.
Stop	The cancel signal fired 2 ms into every request (Node 20), so Stop never stopped and keep-alives never went out.	Stop, or closing the panel, cancels the server work and the model call.
Model status	When the DeepSeek credit ran out, questions just failed.	The account is checked with a free call, at most every 10 minutes; the panel says when the model is down, and /health goes red.
Thumbs up / down	No way to know whether an answer was any good.	Every answer can be rated, with a note. It lands in the audit log.
Same rules as the pages	Late reports 349 against the Operations page's 13; dues followed a different rule from the Money page; the Kidcare branches were left out.	Dues, late reports and branches follow the Money and Operations pages: 13 = 13.
Bugs the benchmark found	Splits silently cut at 200 rows (wrong "most repeating customer"); "CT" read as Clotting Time; "this month" bound to last month; "name him" lost the row.	All fixed. Each question that caught one stays in the benchmark.
A real benchmark	29 invented questions, scored by whether failure flags were absent.	41 of the owner's own questions, typos and all. Truth computed live from the money engine; 8 held out; the judge tests itself (71/71).

Before is September’s production traces of the owner’s real questions. **After** is the new benchmark built from those same questions, run on 7 October against the live database. Every figure below was measured, not estimated.

Measure	Before (Sep)	After (Oct)
Model calls per question, median	7 (mean 11)	2
Steps where the model wrote raw SQL	50%, and 29% of those failed	4%
Most model calls on one question	31	6
Figures typed into the answer by the model	all of them; 48% of answers then needed a rewrite call	none: the server fills them in
Time to answer, median	23 s	6.6 s
Time to answer, 90th percentile	43 s	22 s
“give me cost”	31 calls, 43 s, 13 model-written queries	3 calls, 5 s
“if the CT machine costs me 50 lakh, how long to pay back”	12–16 model-written queries per attempt	5 calls, 22 s
“which doctor is not sending much”	20 calls, 101 s (first October run)	4 calls, 22 s
“read all the reasons for discounts”	12 calls, 53 s (first October run)	6 calls, 28 s
“which test is heavily discounted”	11 calls, 38 s (first October run)	2 calls, 6 s
Collected, commission, CT collected vs the Money page	different formulas, different totals	equal to the rupee (yesterday, month to date, 30 days)
Reports past 24 hours vs the Operations page	349 vs 13	13 = 13
Splits longer than 200 rows	silently cut; ranks wrong	complete
Benchmark, figures and lists	— (29 invented questions)	24/25 (first full pass: 18/24)
Benchmark, open questions	—	7/8
Benchmark, held out (never tuned on)	—	7/8
Cost per question	about \$0.04 (estimated from traces)	under \$0.001 (\$0.03 for all 41, measured)

The three misses in that run. Two were bugs, fixed since and re-run: the “ct” inside “CT-BRAIN PLAIN” was also read as *every CT scan*, losing Rs. 4,400 of orders; and “which reports take the longest” asked for turnaround by test, which has no catalog shape, and was not handed to SQL. The third gave dues by age correctly but never stated their total.

Why it is faster and cheaper at the same time: a catalog step costs no model call and cannot fail to compile, so the plan that used to spend 13 calls writing and repairing SQL now spends none. The latency that is left is the two model round trips.

The three things still wrong in September — where they stand

1. Grain enforced by name. half fixed

Money now carries its grain in code: a payment is shared across a bill’s live tests by price, the same split the Money page uses, so “CT collected” adds up to the total. Free SQL still relies on named repairs.

2. Scope verified backwards. half fixed

For catalog steps, scope is now bound *forward*: the filter the owner asked for is compiled into the step, not looked for afterwards. Free SQL is still checked backwards.

3. No degraded mode. half fixed

Today and This month so far need no model, so the panel still shows the numbers when the model is down, and says it is down. Questions still need the model.

Found in the data along the way

- One bill is marked PAID but still has Rs. 800 open, so it is missing from dues on both Pulse and the Money page.
- One referring doctor is stored as two records, which splits their referrals in two in every ranking.
- The Kidcare branches are now counted, because the Money page counts them. If they are test branches, both should drop them together.

Next	Why	What we expect	How we will know
Every question 3 times	One run hides a model that passes once and fails on the rerun.	At least 95% of questions pass all three runs.	pass-all against pass-any in the benchmark.
Morning digest on WhatsApp	The owner should see the numbers without opening the portal.	One message a day with the same lines as the card.	Meta template approved; delivered and read rates.
Thumbs-down becomes a test	Real failures are the questions worth testing.	Each one reviewed weekly and added with a computed truth.	The benchmark grows and its pass rate holds.
Close the catalog gaps	10 of 70 real questions fit no catalog entry, and turnaround by test and report type are still missing.	Model-written SQL under 5% of steps.	Its share in the benchmark and the production log.
Remember what was said	“The machine costs 50 lakh” and “leave out JGG” must be repeated today.	Follow-ups reuse the figures and filters the owner stated.	New chain cases in the benchmark.
Grain and scope for free SQL	The other half of the first two September problems.	Free SQL carries the same grain and scope as a catalog step.	No named repair left in the code.
Answers with no model	The third September problem, finished.	A figure question answered from the catalog when the model is down.	The benchmark run with the model switched off.

WHAT THE OWNER SHOULD SEE				WHAT WE WILL WATCH			
a figure		2 calls, about 4–6 s		benchmark, pass all 3 runs		95% or more	
an open question		2–6 calls, under 30 s		held out vs the rest		within 5 points	
hard ceiling	60 s and 12 calls, and it says so			model-written SQL		under 5% of steps	
money figures	equal to the Money page			thumbs down		read every week	
when the model is down	the numbers, and a notice			model credit	alert before it runs out		
cost per question	under \$0.001			questions asked per week		should rise	

THE BAR, UNCHANGED, IN ORDER OF WHAT IT COSTS THE OWNER

1. Never present a figure for a wider question than was asked — a centre-wide total labelled “Chintal” is the worst output this system can produce. **2.** Never state a number it cannot trace to a step. **3.** Never report a limitation of Pulse as a fact about the business. **4.** Never present an incomplete investigation as settled. **5.** Never attach a card that adds nothing. **6.** Prefer silence to a confident wrong number; silence is recoverable.

October added one: **7.** A money figure that differs from the Money page is wrong, however it was reached.